Almeida Néo

I'm a Software Engineer !

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Experience

PIT | Lille, France | *Apprenticeship* Aug 2024 – Present

Full-Stack & DevOps Engineer (Apprentice) / Full-Stack Developer & DevOps (Apprentice) / Full-Stack Developer (Apprentice)

Full-Stack & DevOps Engineer | Playard Nov 2025 – Dec 2025

Playard centralises several generative AIs in a single interface. The AI-generated V1 had to be reworked and hardened to move from a promising prototype to a production-ready product.

- Audited the AI-generated V1 and mapped the technical debt
- Restructured the architecture (UI, business logic, API integrations)
- Hardened the key flows and reduced non-deterministic behaviour
- Established conventions, refactored and systematically reviewed changes
- Prepared for industrialization (deployment, configuration, maintainability)

Pit Fabrik Aug 2025 – Present

Pit's in-house DevOps team: cloud infrastructure, environments and continuous delivery on Azure.

- Designed, maintained and evolved CI/CD pipelines on Azure DevOps
- Set up and monitored production, staging and development environments on Azure
- Automated deployments and continuously improved the infrastructure
- Contributed to securing and optimising the DevOps processes

Flutter Mobile Developer | Mouv'Aps Aug 2025 – Sep 2025

Health-support mobile app: tailored, progressive and accessible activity programmes. The challenge: deliver a convincing prototype for the pitch in a very short time.

- Adapted the provided template to the targeted use cases
- Designed clean, consistent screens with subtle micro-interactions
- Built a clickable demo prototype (PoC) for the pitch

Full-Stack & DevOps Engineer | LP Solutions Dec 2024 – Present

Boond, the company ERP, did not cover training management. The challenge: connect LP Solutions to Boond to fill that gap, ensuring reliable and consistent data exchanges between the two systems.

- Took part in designing the Boond connection, a key component of the project: NestJS reverse proxy architecture (routing, URL rewriting, exchange validation)
- Took part in the evolutive and corrective maintenance of LP Solutions
- Contributed to setting up the staging and production environments

Full-Stack Mobile Developer | Cocoricorando Sep 2024 – Nov 2024

Cocoricorando centralises and simplifies access to the information hikers need: digital GPX routes, real-time notifications and smooth navigation in an interface designed for the field.

- Built the Flutter app (UI + state)
- TypeScript-typed API backed by PostgreSQL
- Set up the team Git workflow

Full Stack Developer & DevOps Aug 2024 – Aug 2025
Back to studies through a work-study program to deepen full-stack development on real projects.
Institut Laue-Langevin | Grenoble, France | *Permanent* Aug 2023 – Nov 2023

Full Stack Engineer Aug 2023 – Dec 2023

Hardening one of the reactor safety systems, a project under NDA. Curious why this contract was short? Reach out, the story is worth it!

- Wrote the specifications in agreement with the ASN
- Designed the access-control system: physical doors, Yubikey badges, virtual access
- Migrated and hardened the databases (Oracle procedures)
- Built an Angular 17 CMS and a Quarkus backend
- Set up Kubernetes integration and scoped tasks in Taiga

BetterCallDave | Lille, France | *Internship* Mar 2023 – Jul 2023
Mobile Developer Intern | Universal Music Apr 2023 – Jul 2023
Digster is a music streaming and playlist mobile app. The legacy app had piled up technical debt: the goal of the internship was its full rebuild on a recent React Native version.

- Audited and analysed the existing app
- Full migration to a recent React Native version
- Reworked and modernised the legacy features
- Optimised performance, stability and maintainability

BetterCallDave | Lille, France | *Internship* May 2022 – Aug 2022
Full-Stack Intern | Le Fresnoy May 2022 – Aug 2022
OpenProcess is a back-office for building interactive videos. It lets editors add an interaction layer onto a video (coloured or invisible pop-ups, clickable areas) to create the illusion of interactive videos.

- Designed and built the OpenProcess back-office
- Created the interaction editor: coloured or invisible pop-ups positioned on the video
- Rendered the interactions at playback for an interactive-video feel

Education

Engineering degree in Computer Science | IMT Nord Europe Aug 2025 – Present
Engineering programme as an apprenticeship, still with PIT: working towards the engineering degree by combining courses and real-world missions. **BUT in Computer Science** | IUT A, Universite de Lille Aug 2024 – Aug 2025
Final year of the BUT degree completed as a work-study programme at PIT: back to school after professional experience, applying skills directly in a company. **Bachelor's degree (L3) in Computer Science** | Université de Lille Aug 2022 – May 2023
A year consolidating computer science fundamentals, alongside a second internship at BetterCallDave.
DUT in Computer Science | IUT A, Universite de Lille Aug 2020 – Aug 2022
Two years of fundamentals: algorithms, object-oriented programming, databases and web development, concluded with a first internship at BetterCallDave.

Skills

Front-end: React, React Native, NextJS, Angular, TypeScript, CSS, Tailwind
Back-end: Quarkus, NodeJS, ExpressJS, NestJS, TypeScript, Java, Haskell, Rust
Infrastructure: Git, Docker, Kubernetes, Unix, Azure DevOps, Azure
Design: Photoshop, Figma, Illustrator, After Effects

Programming Languages Stack: Java, TypeScript, Rust, C, C++, Haskell

Projects

- **Random-Background-Generator** – Generates abstract geometric backgrounds ready for LinkedIn (1200×628). From a palette or a single color, it stacks shapes (rectangles, triangles, circles…) driven by Simplex noise for an organic look, then applies an ordered effects pipeline (blur, grain, vignette, posterize…)
  Seed-reproducible, available as a CLI or through a small Flask web UI with live preview.
- **EML-Operator** – Based on arXiv:2603.21852, this prototype implements gradient-based symbolic regression over EML trees (the "master formula"), reproducing the exact recovery experiments from §4.3.
- **Deob-mapper** – Matches readable Java sources to their ProGuard-obfuscated counterparts. You have the original .java files with real class/method names and a decompiled dump with mangled names like fBU.java, aCK.java, this tool figures out which is which.
  Private project.
- **Nin-tcha** – A turtle-themed gacha game where you collect and battle creatures against other players. The goal is to collect monsters, build teams, fight other players, and climb the Elo leaderboard
- **Raytracer** – A raytracer written in Java with two entry points: a CLI version (console only) and a GUI version based on JavaFX, with optimization concerns throughout.
- **Travaux Sisters** – Travaux Sisters is a school group project built around two main features:
  - A community help forum for individuals looking to carry out their own home improvement work (DIY, renovation, etc.) - A quote analysis tool that breaks down and explains technical terms to help non-expert users better understand their quotes
- **NinAnimate** – NinAnimate animates the transition from one piece of code to another, token by token (like a "magic move"): what doesn't change slides into its new position, while the rest fades in and out. It's all incredibly smooth!
  Chain your code steps together like slides, zoom in on what matters, and export the entire animation as a video (MP4)—perfect for presentations, tutorials, or blog posts!
  Inspired by the "magic diff" feature of some plugins and videos seen online.
- **Typescript-Horrors** – For a "Nuit de l'Info" hackathon, one challenge was to build something functional but architecturally… disgusting. So I made a fully recursive, fully type-safe RPN calculator: reverse Polish notation, infix-to-RPN conversion, i18n, custom utility types… and a few crashes. Approved by peers.
- **NinIde** – An IDE written in C, inspired by Vim, with several editing modes, a Ctrl-F search and a few fun features.
- **Wakfuli** – Wakfuli is a builder for the video game Wakfu, designed to help players create and optimize their builds easily. Developed by a team of three enthusiasts, the project aims to provide a smooth and intuitive interface for customizing equipment, spells, and abilities.
  While not open-source for now, our goal is to deliver a powerful tool for the community.
- **NeoSnake** – Technical test for a company called "Pit" They asked for a simple snake.. I found that dull, so I built a competitive snake game that requires skill to master ! Inspired by Trackmania and Happy Wheels, focused on speed and skills.
  Players can also design and share their own levels.
- **Star-Realms-HpCounter** – A companion tool for the Star Realms deckbuilding game. It's a 1v1 between two fleets where you drain your opponent's hit points — tedious to track in your head, hence this small app.
- **Cpu-Emulator** – Emulation of a CPU and ASM parsing (an ALU specifically). Paused and unfinished, but a fascinating project to dig into.
- **3D Engine (old)** – Group project carried out as part of my computer science degree, where the goal was to create a "complex" project. We therefore chose to explore 3D through this project in view of the master's program we wanted to pursue, the Master in Virtual and Augmented Reality.
- **HaskellHorrors** – My university Haskell labs, plus some LeetCode exercises and exams. I loved Haskell — it's what got me into type systems and functional programming, which I've since reused on

other projects.

- **ICAL-Fac** – A Discord bot that displays your university timetable. The faculty offered nothing practical and the official schedule was a pain to read .. this bot finally makes it readable !
- **FractalDisplayer** – This project is a basic fractal viewer that is currently capable of displaying the Mandelbrot fractal. Its purpose is to help me learn and explore the use of shaders.
- **Maze Things** – Simply a maze generator, that can solves himself the maze you provides him
- **Image Processing** – Little python project aimed at facilitating the learning of image processing techniques.
- **Client SSH** – This is an SSH client, implemented using the Java programming language.
- **JustSeries** – Project carried out by a team of 3, with the simple goal of using an API. This website allows users to search for TV series and access certain information such as their ratings, etc.
- **Netcast** – This is a web server that has been implemented using the C programming language.
- **Windows's Notepad copy** – This project is a replica of Windows' Notepad, with additional features. It provided me with the opportunity to explore C# and .NET.
- **3D Modelisation Tool (Only for .ply)** – I've worked with a "group" for this 3D modeling tool, that allows you to open .ply files and view them in a visual format.
- **Ch-hess** – It is a chess game called Ch-hess, inspired by the term "Hess" frequently used by young people to denote being in a state of financial difficulty.
- **ColorFx** – Tool developed with the aim of exploring JavaFX, but also to solve a problem with black and white printing, as colors are often poorly chosen, making handouts unreadable.
- **Sokoban** – A sokoban game developed during my very first year of study, a puzzle game where you have to push boxes into holes, completed as a group project in C. We even chose to use Ncurses to make the task more challenging